SentinelAI

Transforming Every Streetlight into a Silent Guardian

AI-Driven Distributed Public Safety Mesh for Smart Cities — Concept Document

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1. Vision & Mission

Vision. Build an AI-powered public safety infrastructure that transforms existing streetlights and public spaces into intelligent emergency response systems — reducing emergency response time, improving women's safety, and saving lives.

Mission. Leverage Artificial Intelligence, Edge Computing and IoT to create a scalable, affordable, privacy-conscious public safety platform for India, and eventually the world.

2. Problem Statement

Every year, thousands of people lose their lives in road accidents because emergency response arrives too late. Women and vulnerable individuals face safety threats in isolated public spaces where immediate help is unavailable. Current solutions fall short:

- CCTV cameras only *record* incidents — nobody responds in real time.
- Emergency apps require a conscious victim to manually request help.
- Public infrastructure is passive rather than intelligent.
- Emergency services receive delayed, incomplete, unverified information.

India needs a system that can **detect, verify and respond** to emergencies automatically.

3. Solution

SentinelAI transforms existing streetlights, bus stops and public infrastructure into AI-powered emergency response nodes. Each smart node continuously monitors its surroundings using on-device AI and multiple sensors. Instead of waiting for someone to report an incident, SentinelAI automatically **detects** accidents and distress situations, **verifies** emergencies through multi-sensor fusion, **alerts** authorities and family with exact GPS location, **guides** emergency responders along the safest route, and **records** evidence securely.

4. Core Features

AI Accident Detection

Detects vehicle, bike and pedestrian accidents by fusing the AI camera, microphone and vibration sensor.

Women's Safety

Triggered by an SOS press, a scream, or AI-detected suspicious activity. SentinelAI alerts police and family, increases street lighting, activates nearby cameras, and records evidence.

Safe Corridor

Nearby smart poles communicate as a network. During an emergency, poles along the route brighten, cameras hand off tracking, responders receive the safest route, and units know exactly where to go.

Edge AI

Data is processed locally on the pole instead of streaming to the cloud — faster response, lower bandwidth, better privacy.

Predictive AI

Beyond detection, SentinelAI predicts accident-prone roads, unsafe locations, repeated crime hotspots and infrastructure failures.

5. System Components

Hardware (per node)

- AI camera
- Microphone
- Vibration sensor
- GPS module
- SOS button
- Emergency LED light
- Speaker
- Edge AI processor
- IoT communication module
- Backup battery

Software

- **AI:** YOLO, OpenCV, TensorFlow, Edge Impulse
- **Frontend:** React, Tailwind CSS
- **Backend:** Node.js, Express
- **Database:** MongoDB
- **Cloud:** AWS
- **Notifications:** Firebase, SMS gateway
- **Maps:** Google Maps

6. System Architecture

Sensors (camera · microphone · vibration · GPS · SOS button) → **Edge AI Processor** → **Emergency Validation Engine** → **Response fan-out:** Police · Ambulance · Family · Cloud Dashboard · Smart Streetlights

7. User Flows

Accident. Vehicle crash → AI detects impact → microphone confirms sound → vibration confirms collision → AI verifies emergency → GPS locks location → police notified → ambulance notified → streetlights brighten → evidence stored.

Women's safety. SOS pressed → AI activates → nearby cameras start recording → emergency lights ON → police receive live location → family notified.

8. Target Users & Market

Primary beneficiaries	Women, senior citizens, daily commuters, highway travellers, students
Customers	Police departments, Smart City SPVs, municipal corporations, transport authorities, universities, industrial parks
India market	100+ Smart Cities, municipal corporations, state governments, NHAI, metro rail, airports, private townships
Future markets	Middle East, Southeast Asia, Europe

9. Business Model & Revenue Streams

- **Hardware** — smart-node sale and installation (one-time, per node)
- **SaaS** — annual dashboard subscription per city / campus
- **Maintenance** — annual maintenance contracts (AMC)
- **AI analytics** — monthly hotspot and infrastructure reports
- **API licensing** — incident data feed for maps, insurers, fleets
- **Smart City integration** — government project contracts

10. Innovation

Unlike traditional CCTV, SentinelAI **detects, understands, verifies and responds** instead of simply recording. The Safe Corridor feature lets multiple smart poles coordinate an emergency response as a network rather than acting independently.

Patent opportunities

- Multi-sensor emergency validation algorithm
- Distributed AI communication between smart poles
- AI-driven Safe Corridor activation
- Edge-based emergency detection framework

Patentability depends on detailed prior-art searches and the specific technical implementation — a formal search is a Phase 1 deliverable.

11. Roadmap & Prototype Cost

Phase 1	Research & validation — stakeholder interviews, requirements, prior-art search
Phase 2	Prototype — single working node: sensor fusion, edge AI, dashboard, alerts
Phase 3	Campus pilot — small mesh, Safe Corridor demo, false-alarm tuning
Phase 4	City pilot — deployment with a municipal partner
Phase 5	Commercial product — Smart City tenders, SaaS, manufacturing partners

Prototype estimate: ₹18–20 lakh — covering AI cameras and sensors, edge processors, cloud and dashboard build, mobile alerts, and field testing.

12. Risks & Safeguards

Privacy concerns, false alarms, weather effects on sensors, hardware maintenance, government approvals and installation costs are addressed through privacy-by-design (edge-first processing), multi-sensor fusion, local AI, maintenance planning, and partnerships with public authorities.

13. SDG Alignment

Supports SDG 3 (Good Health), SDG 5 (Gender Equality), SDG 9 (Industry & Innovation), SDG 11 (Sustainable Cities), SDG 16 (Peace & Justice).

14. Future Enhancements

Fire detection · flood alerts · air-quality monitoring · road-damage detection · crowd management · disaster response · gunshot detection · missing-person assistance · smart traffic management.

15. Why SentinelAI Matters

SentinelAI isn't another surveillance system. It makes public infrastructure **aware, responsive and connected** — enabling faster emergency response and safer public spaces. By combining AI, edge computing and IoT with infrastructure cities already own, it seeks to reduce preventable deaths and support India's vision of smarter, more resilient cities.

16. Immediate Next Step: Validation

Before building, the concept will be validated with the people who would use it: police personnel, ambulance operators, municipal officials and potential users. Their feedback will refine the problem definition, surface implementation challenges, and reveal which features are truly valuable — this validation is Phase 1 of the execution plan.